PIXELOTECH

We take delight in our work and culture. Present your goals and visions in front of us and watch us pick up your results. Armed with online meetings, love for leads, and discussing trends, we at Pixelotech, consider every pixel of your brief and curate a clear strategy, vision, and vivid experience for you and your end-users e strive to grab new possibilities, not to stutter or trip, but to walk with determination, to walk with passion, and decode the unknown and the known

Who are we

We design digital products
and explore the interaction between people and technology.

Who you should be...

- Creating something from scratch, taking it to the next 100 million users.
- Just finishing your work can't satisfy you, you look at the product as your own. Its successes are your success and its failures are yours. You eat it, sleep it and feel for it and so will go that extra mile to make it succeed without anyone asking you to.
- You are never satisfied with what you already know. You always want to learn more and the best way for you to learn is by doing.
- You should be able to solve problems that come along your path.

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Assignment

Goal

Build a Task Scheduling Fullstack Web Application where users can:

- Create scheduled tasks (e.g., “Send reminder in 10 minutes”)
- View all tasks and their statuses
- Mark tasks as completed
- See tasks automatically executed by a background worker

Requirements

- Use Django (backend) and ReactJS (frontend).
- Implement APIs to:
    - Create a scheduled task
    - List tasks
    - Mark tasks completed
- Use Celery (or RQ) as a worker for delayed execution.
- Store tasks in a database.
- Add basic tests (API and task execution).
- Build an intuitive user interface for this service, ensuring that all functionalities are seamlessly integrated and fully connected to the backend APIs, allowing users to perform all the aforementioned tasks efficiently.
- Use at least one AI tool (ChatGPT or Copilot or any other) to assist you, and document:
    - What you asked AI
    - What code it generated
    - What you accepted, modified, or rejected. and why
- Provide a README explaining:
    - System design
    - Improvements you’d make if given more time
    - What you learned new during the assignment

Bonus (Optional but highly valued)

- Add retry logic for failed tasks.
- Add rate limiting or throttling.

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- Add API authentication (JWT or session).
- Add Dockerfile + docker-compose setup.
- Add metrics or logging enhancements.
- Polished UI
- Add Optimistic UI updates
- Add Polling or auto-refresh

Constraints:

- Use ReactJS v18.3.1+, Python 3.13+, Django 6.0+, and Django REST Framework 3.16+.
- The database must be PostgreSQL.

Deliverables:

- A link to a **Private** Git repository containing:
  - The complete project with clean, well-organized code.
  - A comprehensive .gitignore file.
- A README.md file that includes:
  - Setup and installation instructions.
  - How to run tests and view coverage reports.
  - Any additional notes on implementation details.
  - System design
  - Improvements you'd make if given more time
  - What you learned new during the assignment
- Document explaining AI tool usage as mentioned in the requirement.
- API documentation accessible at /api/docs/ endpoint.
- A demo video showcasing REST APIs via Postman or similar tool and User Experience on the UI (Audio not necessary)

Submit your Github repo (make it **private**) by inviting us (username: "pixelotech") as a collaborator and send your submission on this link https://forms.gle/itSyPLf6RzV6gch96

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